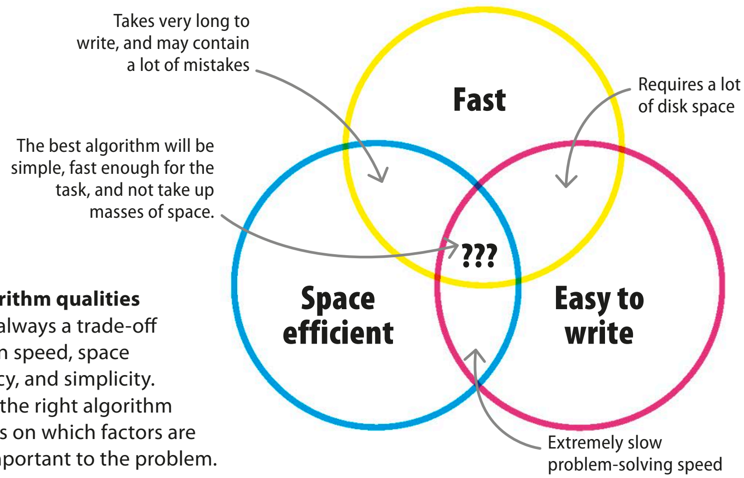
Selecting algorithms

While efficiency is important in an algorithm, there are other factors to consider. First is space efficiency. An algorithm may be quick, but if its speed means it takes up a lot of disk space, it may be better to choose a slower algorithm. The next consideration is how difficult an algorithm is to write. The more convoluted the algorithm, the more potential there is for human mistakes. An algorithm is useless if it's fast and space efficient but gives the wrong answers because bugs have crept into the code.



▷ **Algorithm qualities**
There's always a trade-off between speed, space efficiency, and simplicity. Picking the right algorithm depends on which factors are most important to the problem.

IN DEPTH

Cryptography

Every time secure data is sent over the internet, the message is encrypted by a special algorithm that restricts who can read it. Hackers try to break the encryption to steal private data. Security companies constantly develop new encryption algorithms to stay a step ahead of hackers. Of course, safer algorithms are often slower and harder to write.

Tailoring algorithms

Since inventing new algorithms takes years of study, most developers only implement existing ones. To create a GPS system, developers will have to model the data (roads, cars, traffic lights) in a way that the algorithm understands, and then make adjustments for one-way streets, school zones, and toll routes.

▽ **Dijkstra's algorithm**
Dutch scientist Edsger Dijkstra invented an algorithm for finding the shortest path between two points. Variants of his algorithm are used by social media companies to suggest new friend connections to a user, depending on common friends, location, similar interests, and so on.